

Department of Mathematics, IIT Madras
MA2031 Linear Algebra July - November 2023
Assignment 2

1. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove the following:

- (a) A is a subspace of $\mathbb{F}^n$ iff $A = \text{span}(A)$.
- (b) $\text{span}(A \cup B) = \{u + v : u \in \text{span}(A), v \in \text{span}(B)\}$.
- (c) $\text{span}(A \cap B) = \text{span}(A) \cap \text{span}(B)$.
- (d) $A \cup B$ is linearly independent iff $\text{span}(A) \cap \text{span}(B) = \{0\}$.

2. Find bases for the following subspaces of $\mathbb{R}^5$:

- (a) $\{(a, b, c, d, e) \in \mathbb{R}^5 : a + b - c = b + d - 2e = 0\}$
- (b) $\text{span}\{(1, -1, 0, 2, 1), (2, 1, -2, 0, 0), (0, -3, 2, 4, 2), (3, 3, -4, -2, 1), (5, 7, -3, -2, 0)\}$

3. Let $A \in \mathbb{F}^{n \times n}$. Let $\{v_1, \dots, v_n\}$ be a basis of $\mathbb{F}^{n \times 1}$. Prove that A is invertible iff $Ax = v_i$ is consistent for each $i \in \{1, \dots, n\}$.

4. Let $A = \begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ 1 & 1 & 1 & 2 & 1 \\ 5 & 8 & 5 & 7 & 5 \end{bmatrix}$. Write $r = \text{rank}(A)$ and $k = \text{null}(A)$.

- (a) Find bases for $R(A)$ and $N(A)$.
- (b) Express suitable $4 - r$ rows of A as linear combinations of other r rows.
- (c) Express suitable $5 - r$ columns of A as linear combinations of other r columns.

5. Let A and B be matrices such that $\text{RREF}(A) = \text{RREF}(B)$. Then, which of the following can be concluded?

- (a) $R(A) = R(B)$. (b) $N(A) = N(B)$.
- (c) The row space of A is equal to the row space of B .

6. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. Consider the ordered bases $B = \{[1, 1, 1]^t, [1, 2, 1]^t, [1, 1, 0]^t\}$ and $C = \{[1, 0, 1]^t, [0, 1, 1]^t, [1, 1, 0]^t\}$. Compute $[A]_{C,B}$, $[A]_{B,C}$ in the following ways:

- (a) By expressing A -images of the basis vectors in B in terms of basis vectors of C , and vice versa.
- (b) By computing the inverses of the matrices constructed from the vectors in both bases.
- (c) By using the RREF conversion of suitable augmented matrices.

7. Let $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 1 & 3 \\ -1 & -1 & -2 \end{bmatrix}$, $V = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a + 2b + 3c = 0, a, b, c \in \mathbb{R} \right\}$,
 $B = \{[0, 3, -2]^t, [2, -1, 0]^t\}$ and $C = \{[1, 1, -1]^t, [3, 0, -1]^t\}$.

- (a) Show that both B and C are bases for V .
 - (b) Extend B to a basis O of $\mathbb{R}^{3 \times 1}$.
 - (c) Extend C to a basis N for $\mathbb{R}^{3 \times 1}$.
 - (d) Show that $A : V \rightarrow V$ given by $A(v) = Av$ is well defined.
 - (e) Find the change of basis matrix $I_{N,O}$.
 - (f) Find the coordinate matrix $[A]_{N,O}$.
 - (g) Verify that $[v]_N = I_{N,O}[v]_O$ and $[Av]_N = [A]_{N,O}[v]_O$ for $v = [-4, -1, 2]^t$.
8. Find a full rank factorization of the matrix A of Question No. 4.
9. Which are the matrices similar to, and equivalent to the
- (a) zero matrix? (b) Identity matrix?
10. Using the rank factorization, prove the following:
- (a) The row rank of a matrix is equal to its column rank.
 - (b) If AB is well defined, then $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.
11. Is $\{(1, 2, 3, -1), (2, -1, 0, 0), (0, 0, 1, 3)\}$ an orthogonal set in $\mathbb{F}^3$?
12. Construct an orthonormal set from $\{(1, 2, 0), (2, -1, 0), (0, 0, 2)\}$.
13. Given two linearly independent vectors in $\mathbb{F}^3$, how to construct a third vector so that these vectors form a basis for $\mathbb{F}^3$?
14. Let $\{u, v\}$ be an orthonormal basis for $\mathbb{R}^{2 \times 1}$. Let $x \in \mathbb{R}^{2 \times 1}$ be a unit vector. If $x^t u = 1/2$, then what could be $x^t v$?
15. Find QR-factorization of the following matrices:
- (a) $\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$
 - (b) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$
 - (c) $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
16. Let $\{u, v, w\}$ be an orthonormal basis for a subspace V of $\mathbb{R}^{5 \times 1}$. Let $x \in V$ with $\|x\| = 5$, $\langle x, u \rangle = 4$ and $x \perp v$. Can you express x as a linear combination of u, v, w ?
17. Find the best approximation of $(1, 2, 1)$ from $V = \{(a, b, c) \in \mathbb{R}^{3 \times 1} : a + b + c = 0\}$.
18. Find least squares solution(s) of the system of linear equations:
 $x + y + z = 0, -x + z = 1, x - y = -1, y - z = -2$.